CSD415 PROJECT PHASE – I

Extended Abstract

Behavioral Biometric-Based Continuous Authentication System for Secure Mobile Banking Using Machine Learning

Team Members

6	Aiswarya R	IDK23CS006
44	Niranjana K D	IDK23CS045
61	Vimal Vijayan	IDK23CS063
62	Vishnu P S	IDK23CS064

Project Guide

Dr. Dhanya S Pankaj

Department of Computer Science and Engineering
Government Engineering College Idukki

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1 Problem Statement

To develop a multimodal behavioral biometric-based continuous authentication system for secure mobile banking sessions using machine learning.

Mobile banking applications today authenticate a user only once, at login, through mechanisms such as passwords, PINs, One-Time Passwords (OTPs), fingerprints, or facial recognition. This leads to several critical shortcomings that motivate the present work:

- Once a session begins, the system assumes that the same individual continues to operate the device, with no further identity checks performed until logout or timeout.
- If a device is lost, stolen, left unlocked, or its session is hijacked after login, an unauthorized individual can freely operate the banking application – transferring funds, viewing sensitive data, or altering account settings – without further verification.
- This risk affects both banking customers, who are exposed to fraud, and financial institutions, which bear liability, regulatory scrutiny, and reputational damage.
- Existing single-check and static-biometric approaches verify identity only at one point in time and force a trade-off between security (short session timeouts) and usability (long timeouts).

Example: A user unlocks their phone with a fingerprint and leaves it unattended – the system has no way of recognizing that a different individual is now operating the session.

Problem Illustration

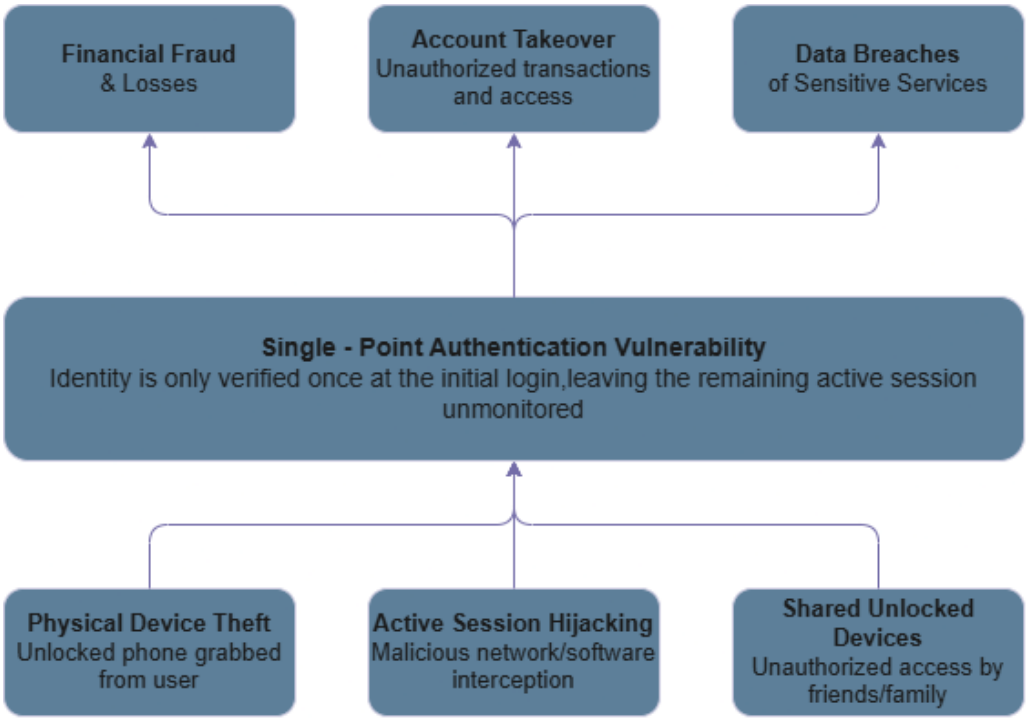


Figure: Problem illustration showing how single-point authentication vulnerabilities (device theft, session hijacking, and shared unlocked devices) lead to financial fraud, account takeover, and data breaches.

2 Motivation and Need

- **Social Relevance:** Protects banking customers from financial fraud without imposing repeated authentication prompts that disrupt the user experience.
- **Industrial Relevance:** Aligns with the growing regulatory push toward zero-trust security models and frictionless fraud prevention in banking and fintech.
- **Economic Benefit:** Requires no additional hardware, keeping deployment cost low while reducing fraud-related losses and chargebacks for financial institutions.
- **Environmental Benefit:** Entirely software-based, with no additional physical sensors or devices to manufacture or deploy.
- **Need for Automation:** Replaces manual, disruptive re-authentication with passive, machine-learning-driven trust evaluation that operates transparently in the background throughout the session.

3 Objectives

Overall Objective

To design and implement a multimodal behavioral biometric-based continuous authentication system that passively verifies user identity throughout a mobile banking session using machine learning, and dynamically manages session trust without disrupting the user experience.

Specific Objectives

1. Curate a multimodal behavioral dataset (keystroke dynamics, touch/swipe gestures, and device motion) collected during simulated mobile banking interactions.
2. Design and extract discriminative behavioral features for each modality, such as keystroke dwell/flight time, swipe velocity and pressure, and accelerometer/gyroscope motion patterns.
3. Implement and comparatively evaluate anomaly detection models – One-Class SVM, Isolation Forest, and Autoencoder-based architectures – to detect deviations from a legitimate user's behavioral profile.
4. Design a dynamic trust-scoring mechanism that fuses multimodal anomaly scores to determine whether to continue the session, request step-up authentication, or terminate access.
5. Evaluate system performance using False Acceptance Rate (FAR), False Rejection Rate (FRR), Equal Error Rate (EER), Precision, Recall, F1-Score, and authentication latency.
6. Develop a mobile application prototype with a supporting API to demonstrate real-time continuous authentication in a simulated banking scenario.

4 Literature Survey

At least ten recent research papers were reviewed to identify existing approaches, their limitations, and the gap this project addresses. Table 1 summarizes the comparison.

Table 1: Comparison of reviewed literature on behavioral biometric continuous authentication.

Paper	**Method**	**Dataset**	**Result**	**Limitations**
do Nascimento et al. (2024) [1]	Random Forest, KNN, and Support Vector Classifier on touch/gesture data	15 users, Samsung tablet, 15 minutes each	SVC achieved ∼90% average accuracy	Small user pool; single-application context; RF showed overfitting
Hu et al. (2025) [2]	Lightweight masked one-class autoencoder trained only on legitimate-user data	Smartphone touch/sensor behavioral data	Effective unsupervised detection without attacker samples	One-class training limits exposure to real attacker behavior; added model complexity
Abba et al. (2025) [3]	Multimodal framework: keystroke, mouse movement, contextual risk factors	3,600 samples, 24 users (public + synthetic)	98.25% accuracy, 0% EER, AUC-ROC 1.0000, latency <2.6 ms	Synthetic data reliance; healthcare/remote-work tuning may not transfer to banking
Acien et al. (2022) [4]	LSTM-based recurrent network for free-text keystroke verification	Large-scale keystroke dataset (100,000+ users)	Scalable deep-learning verification with low error rates	Keystroke-only modality; requires sufficient typing volume
Lee and Chen (2025) [5]	Keystroke-dynamics-driven trust protocol with continual key updates	Formal protocol analysis and experiments	Robust continuous trust evaluation with key management	Protocol-focused; keystroke-only; limited classification benchmarking

Paper	Method	Dataset	Result	Limitations
Stragapede et al. (2022) [6]	Multimodal passive authentication (touch + motion)	Large-scale mobile sensor dataset	Multimodal fusion improves robustness	Environmental noise (walking, grip variation) affects stability
Lin et al. (2023) [7]	Cross-scenario keystroke feature adaptation	Multi-scenario keystroke dataset	Improved consistency across typing scenarios	Keystroke-only; adaptation overhead
Kiyani et al. (2020) [8]	Recurrent confidence model with ensemble learning	Public keystroke dynamics dataset	Improved robustness via ensemble modeling	Older baseline; single modality
Massoli et al. (2022) [9]	Multilayer one-class deep classification (MOCCA)	General anomaly detection benchmarks	Improved one-class accuracy over single-layer baselines	Not biometrics-specific; needs domain adaptation
Wu et al. (2020) [10]	Deep one-class neural network for anomaly detection	Complex-scene video anomaly benchmarks	Strong one-class deep learning baseline	Designed for visual anomalies, not behavioral biometrics

Research Gap

- Most existing systems rely on a **single modality**, most often keystroke dynamics, which fails when the user is not actively typing – for example, while browsing balances or navigating menus.
- Touch- and gesture-based systems, while promising, have generally been validated on **small user samples** and in **non-banking contexts** such as mobile games.
- One-class autoencoder approaches effectively solve the problem of missing attacker data but are typically tested on a **single modality or generic sensor streams**, not a fused multimodal profile.
- **What this project improves:** Combines typing, touch, swipe, and device-motion signals into a single per-user behavioral profile, evaluated through comparative one-class anomaly detection models, and driven by a **graduated, dynamic trust score** – rather than a single binary accept/reject decision – tailored specifically to real-time mobile banking sessions.

5 Proposed Solution

- The system passively captures behavioral signals – keystroke timing, touch/swipe patterns, and device motion – in the background as the user interacts with the banking application.
- These signals are converted into structured feature vectors and compared against the user's enrolled baseline profile using anomaly detection models.
- A continuously updated **trust score** reflects how closely ongoing behavior matches the established profile.
- If the score falls below a defined threshold, the system escalates its response – prompting secondary authentication, or terminating the session outright if the deviation is severe.
- This maintains strong security while requiring no constant active input from the legitimate user.

Architecture Diagram

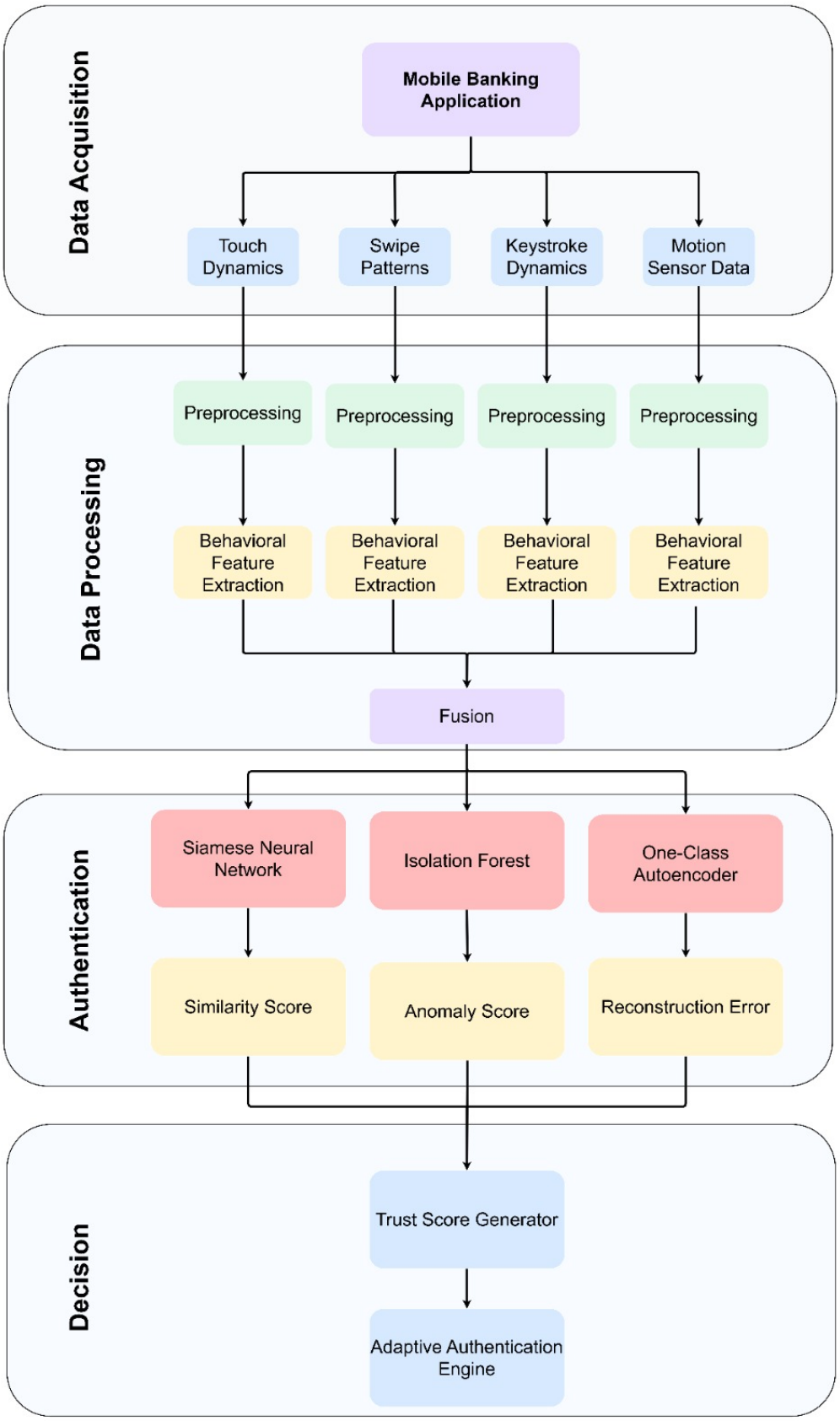


Figure 1: System architecture showing the Data Acquisition, Data Processing, Authentication, and Decision layers of the proposed continuous authentication framework.

6 Methodology

- **Algorithms/ML Models:** One-Class SVM, Isolation Forest, Autoencoder-based anomaly detection, and a trust-score fusion layer combining per-modality outputs.
- **Programming Languages:** Python for model development; Kotlin or Flutter/React Native for the mobile application; Node.js/Flask/FastAPI for the backend.
- **Software:** scikit-learn, TensorFlow or PyTorch, Android Studio, PostgreSQL/Firebase, Postman for API testing.
- **Hardware:** Android smartphones for data collection and testing; a development workstation for model training.
- **Frameworks:** REST API for third-party integration, with optional on-device inference via TensorFlow Lite to minimize latency during continuous checks.

Workflow Diagram

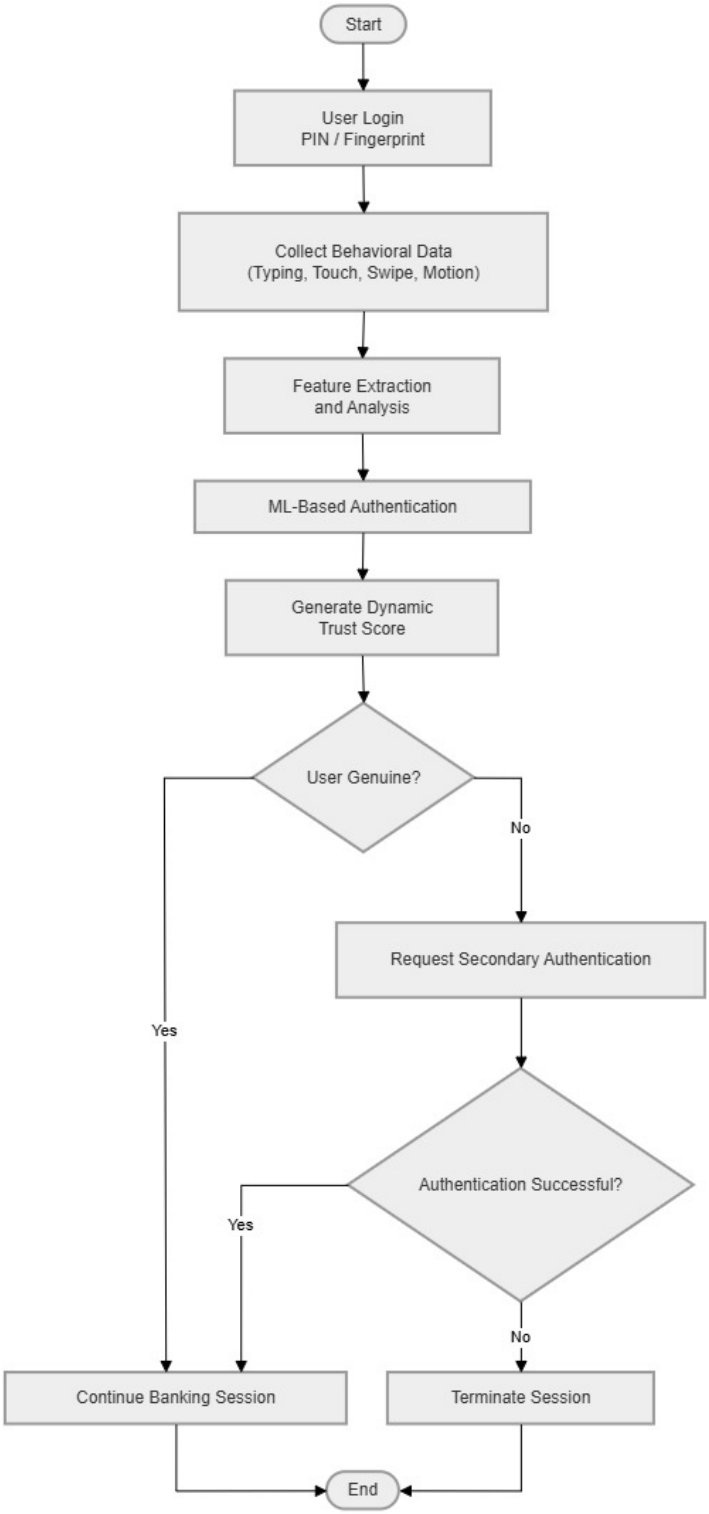


Figure 2: Workflow of the proposed system: from user login and behavioral data collection through feature extraction, ML-based authentication, dynamic trust scoring, and the final session decision.

7 Modules

1. **Data Acquisition Module** – Captures raw keystroke timing, touch/swipe coordinates and pressure, and device motion data in real time.
2. **Feature Engineering Module** – Converts raw behavioral streams into structured feature vectors per modality.
3. **Behavioral Profiling and Anomaly Detection Module** – Trains and executes per-user one-class models to score deviation from the enrolled profile.
4. **Trust Score and Decision Engine Module** – Aggregates multimodal anomaly scores into a unified trust score and triggers the corresponding session action.
5. **Mobile Application and API Module** – Provides a user-facing banking application prototype and a REST API for third-party integration.

8 Resources Required

Software

- Python, scikit-learn, TensorFlow/PyTorch
- Android Studio or Flutter SDK
- Flask/FastAPI, PostgreSQL/Firebase

Hardware

- Android smartphones for data collection and testing
- Development laptop with GPU support for model training

Dataset

- **Name:** Self-collected multimodal behavioral dataset (working title to be finalized)
- **Source:** Volunteer users interacting with a prototype banking-style application (login, fund transfer simulation, balance inquiry, navigation)
- **Number of Samples:** To be finalized based on the number of participating volunteers and sessions per user
- **Features:** Keystroke hold/flight times; swipe coordinates, velocity, pressure, and duration; accelerometer and gyroscope readings

9 Expected Outcomes

- A functioning continuous authentication engine with measurable FAR, FRR, and EER benchmarked against reviewed literature.
- Improved usability through reduced forced re-authentication compared to fixed session timeouts.
- A working mobile application prototype demonstrating real-time behavioral trust scoring.
- A reusable API enabling integration of continuous authentication into third-party banking and fintech platforms.

10 Innovation

- **What is new:** A fused, multimodal behavioral profile combining typing, touch, swipe, and device motion into a single, continuously updated trust score.
- **How it is different:** Uses a graduated response mechanism (continue / step-up / terminate) instead of a binary accept/reject decision.
- **Contribution:** Adapts recent one-class anomaly detection techniques specifically to a live mobile banking session context, validated through a working mobile application and API rather than offline evaluation alone.

11 Project Timeline

Task	Aug	Sep	Oct	Nov
Literature Survey	✓	✓		
Dataset Collection		✓	✓	
Design		✓		
Implementation			✓	✓
Testing				✓
Documentation			✓	✓

Table 2: Gantt chart showing the planned project timeline.

12 Team Work Distribution

The project work is divided across four members, each leading one core module while contributing to integration and documentation as a team. Table 3 summarizes the re-

sponsibility split.

Table 3: Team work distribution across project modules.

Member	Responsibility
Member 1 Data Acquisition & Dataset Lead	Design and implement the data capture module for keystroke, touch/swipe, and motion sensor data.
	Coordinate volunteer recruitment and multimodal dataset collection sessions.
	Perform data cleaning, labeling, and organization of the dataset for training and testing.
Member 2 Feature Engineering & Preprocessing Lead	Design the preprocessing pipeline for each behavioral modality.
	Extract and engineer features (keystroke dwell/flight time, swipe velocity/pressure, motion statistics).
	Perform exploratory data analysis and feature selection/validation.
Member 3 ML Modeling & Evaluation Lead	Implement and tune One-Class SVM, Isolation Forest, and Autoencoder anomaly detection models.
	Design and implement the trust-score fusion and decision engine logic.
	Evaluate models using FAR, FRR, EER, Precision, Recall, F1-Score, and latency; conduct comparative analysis.
Member 4 Application & Integration Lead	Develop the mobile banking application prototype (UI/UX, session handling).
	Build and integrate the REST API connecting the mobile app with the backend ML engine.
	Conduct system testing and prepare documentation, the user manual, and presentation materials.

Note: All members jointly contribute to the literature survey, overall system design, and final report preparation.

13 Expected Deliverables

- Software Prototype
- Mobile Application
- Research Paper
- User Manual

14 Keywords

Continuous Authentication, Behavioral Biometrics, Machine Learning, Anomaly Detection, One-Class Classification, Zero Trust, Mobile Banking Security, Keystroke Dynamics

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